

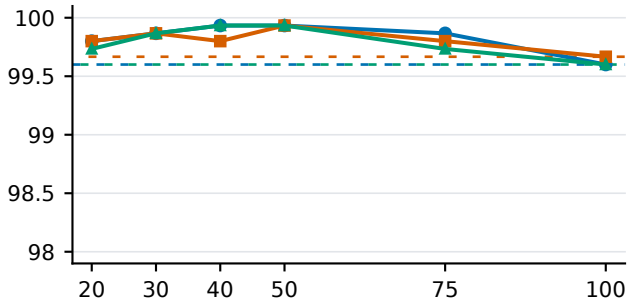
RULER 4K: retrieval subtypes

Graph-based selector FastKVzip KVzip

Dashed: full cache (method color; gray if shared). Y-axis range varies by panel.

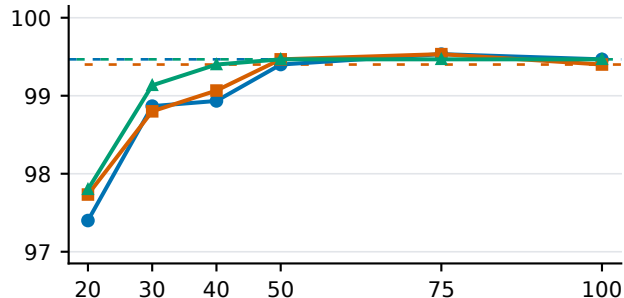
Single NIAH

3 of 3 configurations evaluated



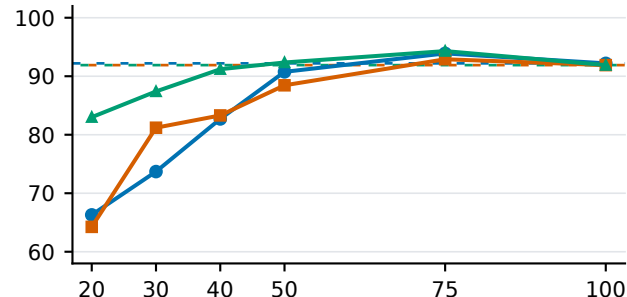
Multi-keys NIAH

3 of 3 configurations evaluated



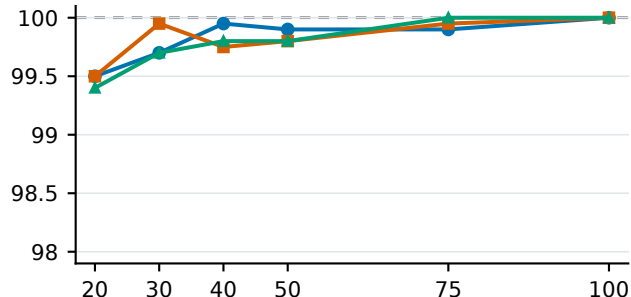
Multi-values NIAH

1 of 1 configurations evaluated



Multi-queries NIAH

1 of 1 configurations evaluated



Retention ratio (%)

Categories follow RULER Section 3. Single and multi-key means combine their available variants. A one-configuration family reproduces its individual task curve.